



BALF Research with Raman Spectroscopy

The development of extremely sensitive CCD detectors and holographic-notch filters is fueling a resurgence in the field of Raman spectroscopy, particularly in the study of organic structures. Traditionally, using Raman spectroscopy to investigate living systems was hampered in two ways: its inability to detect very low-level signals ($<10^{-9}/10^{-12}$) from the excitation source, and the high stray-light levels in monochromators that masked low-frequency components in proximity to excitation wavelengths. With the more powerful and precise equipment available today, Raman spectroscopy has overcome these problems and is well suited even for the study of complex organic compounds. This note describes the landmark use of Raman vibrational spectroscopy to compare bronchoalveolar lavage fluid (BALF) of normal and alveolar proteinosis (AP) lungs.

Introduction

Raman spectroscopy is a powerful and widely used technique for identifying and quantifying materials based upon their molecular and crystal lattice bond vibrations. It is extremely sensitive, noninvasive, and unaffected by aqueous matrices. Modern technology has taken this technique even further. High-performance CCD cameras, spectrometers, and filters deliver single-photon sensitivity, precisely focused high-energy wavelength emissions, and greatly reduced stray light or radiation, thereby bringing a unique level of flexibility into experimentation.

Southern BioGene, located in Birmingham, Alabama, recognized the potential of using Raman spectroscopy for studying the intramolecular and intermolecular interactions involved in a bilayer lipoprotein matrix. In his September 1999 article, "Structural Changes in the Lipoprotein Complex of Bronchoalveolar Lavage Fluid Detected by Raman Spectroscopy" (Spectroscopy, vol. 14, #9, 21-24), Nihal S. De Silva discusses their groundbreaking use of this technique to measure spectra from bronchoalveolar lavage fluid (BALF) obtained from normal and alveolar proteinosis (AP) patient lungs. Their research, described in the following sections, shows that with the right equipment, Raman spectroscopy is an effective means for studying complex living systems.

Application Background

The surfactant lipoprotein system of the lung is made up of roughly 90% lipids (the majority of which are phospholipids) and 5 to 10% specific proteins (the apoproteins SP-A, SP-B, SP-C, and SP-D). The cells secrete this mixture of lipids and proteins into the alveoli and respiratory air passages, where it is transformed into the tubular myelin lattice (TML). The TML forms a coating film that reduces the surface tension of pulmonary fluids and contributes to the elastic properties of pulmonary tissue, thereby limiting collapse during inhalation and exhalation.

The goal at Southern BioGene was to investigate how the biochemical structure of the lipid-protein complex facilitates host defense and surfacetension reduction at a molecular level in the TML. The development of replacement surfactants for respiratory therapy depends on understanding this process. Southern BioGene examined the lipoprotein complex of TML and multilamellar structures, paying particular attention to the interaction of the lung apoproteins SP-A, SP-B, and SP-C with defined phospholipids. Using Raman spectra obtained with the equipment and methods detailed below, they compared the BALF from normal lungs to that of AP patients to investigate possible differences in constituent apoproteins and phospholipid makeup.



Research Equipment and Methods

Southern BioGene has put together a state-of-the-art registration system based on the Acton Research SpectraPro® 2150 f/4 spectrograph, with a reciprocal linear dispersion of 5 nm/mm for 1200 g/mm grating, and a thermoelectrically cooled 1024 x 256 CCD camera (the Princeton Instruments SpectruMM:GS256 detector - predecessor of the modern PIXIS platform). The CCD has a 7 x 25-mm imaging area with 24 x 24- μm pixels. By using a set of gratings with different groove densities (600 and 1200 g/mm), they were able to cover both 60- and 125-nm spectral windows. A grating with a low groove density was utilized to get a full Raman spectrum at ≤ 2600 wavenumbers relative to the excitation wavelength. To analyze different regions of the Raman spectrum with a high spectral resolution ($\leq 4\text{ cm}^{-1}$), a grating with a high groove density was used. A Pentium-based personal computer controlled the spectrograph and the CCD and collected and analyzed the experimental data.

The laser-excitation source was a TEM₀₀ linearly polarized HeNe laser ($\lambda = 632.8\text{ nm}$) with 10-mW output power (Uniphase, San Jose, CA). For sample excitation, the team from Southern BioGene placed the laser 5 m away from the sample and used first-order reflection of the laser beam from an 1800-g/mm diffraction grating. Several apertures were used to control the beam path. These steps were taken to clean up the spectrum of the HeNe laser and to prevent spectral lines of He and Ne gas discharge in the laser tube from entering the spectrograph. As a result, the spectral region of interest was virtually clear of parasitic lines. The output power of the first-order reflected beam was about 3 mW at 632.8 nm. Thermal effects during sample excitation were avoided by using low-power excitation. This also prevented thermal damage of the sample.

Southern BioGene investigated two BALF samples collected from AP patients and healthy subjects at the University Medical Center Hospital in Birmingham, Alabama. The samples were extracted from the lungs with 0.9% NaCl (saline) and stored at -20°C until use. Samples were thawed at 4°C for 2 hours to allow compaction of the multiple layers. Immediately before analysis, each sample was carefully mounted in the Raman setup. The cuvette was positioned as shown in Figure 1 to prevent both the stray reflection of light from the cuvette's top wall and the unwanted Raman signal from the water layer.

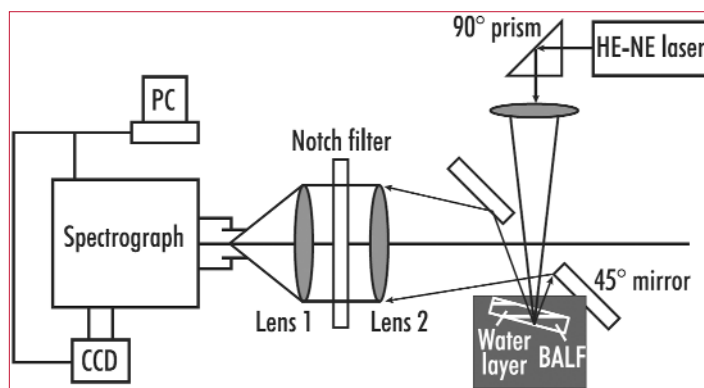


Figure 1: Schematic drawing of Southern BioGene's experimental arrangement to measure Raman spectra of BALF.

The laser beam was focused into the sample by a 500-mm lens through a hole in a 45° mirror. Using a back-scattered scheme to collect the Raman signal, the scattered radiation was reflected by the mirror and then focused by a two-lens objective ($f/1 = 50\text{ mm}$, $f/2 = 75\text{ mm}$) onto the input slit of the spectrograph. A holographic-supernotch filter with an optical density of >6 at 632.8 nm (Kaiser Optical Systems, Ann Arbor, MI) was employed to prevent the excitation wavelength from entering the spectrograph. The spectral bandwidth of the supernotch filter was $\sim 350\text{ cm}^{-1}$. Southern BioGene also used a high-power alexandrite laser tunable in the 730 to 810-nm range, an ideal range for Raman excitation of biological systems. Excitation by alexandrite laser with $\lambda_{\text{ex}} = 785\text{ nm}$ exhibited sufficient signal-to-noise gain of the Raman spectra.



Research Results

The data collected and analyzed by Southern BioGene were the results of careful subtraction of BALF extractant (0.9% NaCl in distilled water) and quartz (cuvette wall) Raman spectra. The specific frequency changes in the spectra of AP and normal BALF samples are readily apparent in Figures 2 and 3.

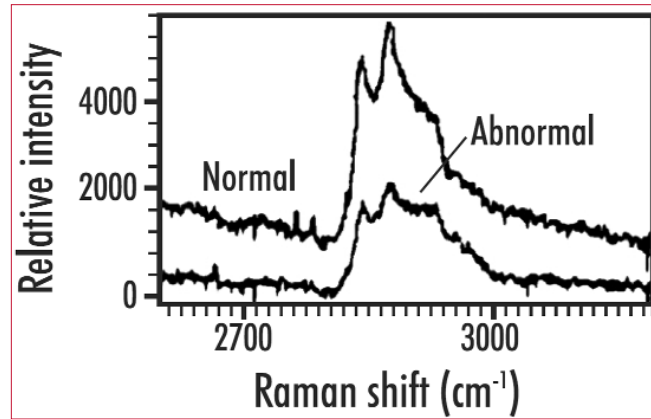


Figure 2: Raman high-frequency spectra. Normal is BALF from control patients and abnormal is BALF from AP patients.

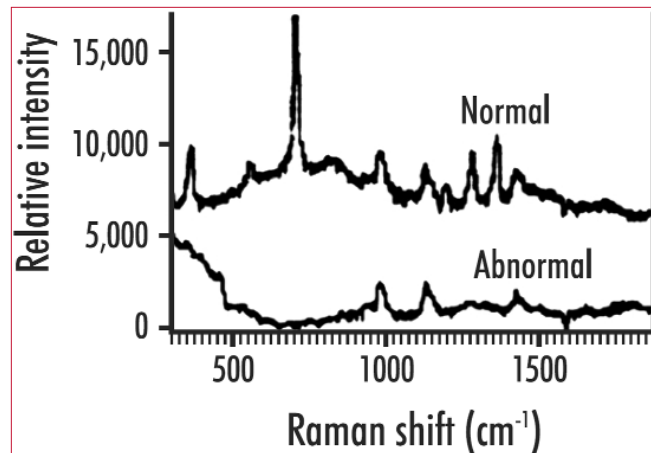


Figure 3: Raman mid-frequency spectra. Normal is BALF from control patients and abnormal is BALF from AP patients.

The high-frequency area of the Raman spectra contains the lipid acyl chain methylene and methyl C-H stretching modes. The Raman band 2850 cm^{-1} refers to the acyl chain methylene's (CH_2) symmetric stretching vibrations, whereas 2880 cm^{-1} refers to its asymmetric stretching vibrations. The 2935-cm^{-1} feature represents a complex interval that contains spectral components from Fermi resonance interactions involving the chain methylene moieties, and separately the C-H symmetric stretching modes of the lipids' chain methyl termini. The spectral intensity ratio $m = I_{2850}/I_{2880}$ shows pure lateral chain-chain interactions. The intensity ratio $n = I_{2935}/I_{2880}$ reflects chain-chain interactions as well, but these also contain contributions from intrachain trans/gauche isomerizations. Southern BioGene's results show that the m value is the same (0.83) for both the AP and normal samples, whereas the n values are significantly different (0.78 and 0.56, respectively). De Silva believes this indicates a similar lateral chain-chain ordering for the samples, but a different intrachain structure and, presumably, a different degree of perturbation in the various portions of the bilayer.



BALF Research with Raman Spectroscopy

In the mid-frequency region, the absence of Raman lines 380-, 730-, 1220-, 1300-, and 1380 cm^{-1} from the AP BALF sample indicates a difference in constituent structure of diseased and normal lungs. De Silva theorizes that this could be due to the absence of functional groups specific for peptides and amino acids rather than for lipids. These data support the findings of Southern BioGene's earlier electron microscopy studies comparing the BALF structures of normal and AP patients. Based on these results, their conclusion is that the TML from AP patients contains no membrane interdigitization and is poorly developed in comparison to normal lungs. Southern BioGene's use of Raman spectroscopy to show structural changes in constituent apoproteins and phospholipids that occur in the BALF of AP patients is an important step in understanding the molecular structure of TML. Now that the technology exists, the biggest roadblock remaining is simply the relatively few Raman spectra of BALF studies that have been performed. According to De Silva, Southern BioGene is in the process of adding to these preliminary results to obtain meaningful and reproducible data for more detailed conclusions. As researchers like those at Southern BioGene continue to use Raman spectra to study complex living systems, correlative data will begin to amass. This in turn will fuel the development and success of future drug and therapy design.

Components for Raman Spectroscopy

In addition to the spectrograph and CCD detector used in the research above, Princeton Instruments offers a wide range of high-performance spectrometers, detectors, and spectroscopy accessories designed for Raman. The Acton Research SpectraPro monochromators and spectrographs feature automated multiple-grating turrets and a 32-bit microprocessor scanning controller with automated self-calibration. The digital cameras combine exclusive front- or back illuminated CCDs, deep-depletion CCDs, ultra-low-noise amplifiers, and a choice of thermoelectric or cryogenic cooling to provide unparalleled sensitivity. Princeton Instruments also offers complete analytical Raman packages such as single MonoVista and triple TriVista systems. For instance, TriVista system gives researchers the ultimate Raman flexibility providing a simple software switch between Additive and Subtractive modes and allowing using the instrument in Single, Double or Triple configurations. On the other hand, the MonoVista is a dedicated Confocal Raman instrument for life science researchers. Our Raman systems are based on the renowned SpectraPro series research-grade spectrographs featuring proprietary optics coatings for the highest possible throughput over the wide UV-NIR range and the proven rigid design providing excellent stability and spectral reproducibility. The systems can be customized and equipped with the broad choice of CCD and ICCD detectors such as PIXIS, SPEC-10 and PI-MAX platforms providing the highest quantum efficiency curves and the lowest noise electronics. This combination gives you the best signal-to-noise ratio and greatly reduced exposure time. A full line of light sources, sampling accessories, and Confocal microscopes is also available from Princeton Instruments.



www.piac-ton.com

email: moreinfo@piac-ton.com

USA +1.877.4 PIAC-TON | Benelux +31 (347) 324989

France +33 (1) 60.86.03.65 | Germany +49 (0) 89.660.779.3

UK +44 (0) 28.38310171 | Asia/Pacific +65.6293.3130

China +86 135 0122 8135 | Japan +81.3.5639.2741